Gameberry Labs: Enabling uninterrupted gaming experiences to win new players

Tell us your challenge. We're here to help.

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Gameberry increa
user base sixfold
COVID-19 pander
improves player e
by focusing on da
insights provided
BigQuery.

About Gameberry

Founded in 2017, Gameberry Labs is the game company behind mobile games Ludo Star and Parchisi Star. The games are inspired by a traditional Indian board game Parchisi, with a twist for a more universal appeal. Parchisi Star is popular with players in Spain, Colombia, and Panama, whereas Ludo Star has a larger following in India, Pakistan, and the Middle East. With 10 million daily active users in 2020, the company is on its way to meet its mission of 50 million daily active users and beyond.

Google Cloud results

- Delivers a consistent game experience with timely log metrics and snapshots to quickly deploy servers
- Stores 80 TiB of data on Cloud Storage to get insights on user behavior
- Delivers clear and insightful data with Looker Studio

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Industries: Gaming

Location: India

Google Cloud (https://cloud.google.com/)

BigQuery (https://cloud.google.com/bigquery/)

Cloud Storage (https://cloud.google.com/storage/)

Compute Engine (https://cloud.google.com/compute/)

Operations (https://cloud.google.com/products/operations)

Logging (https://cloud.google.com/logging/)

Google Marketing Platform (https://marketingplatform.google.com/about)

Looker Studio (https://marketingplatform.google.com/about/data-studio/)

More than three billion (https://www.dfcint.com/product/video-game-consumer-segmentation-2/)

video game lovers are enjoying their favorite games on consoles, PCs, or mobile devices. In fact, global player spending in mobile games has increased by 27% year-over-year (https://sensortower.com/blog/mobile-games-covid-19-impact) in the second quarter of 2020 to $19.3 billion amid the spread of COVID-19. Following this trend, Gameberry Labs, the mobile game developer for popular board games Ludo STAR and Parchisi STAR, saw its daily active users jump from two million in 2019 to 10 million in May 2020.

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> *—Vinesh Battula, Software Developer,*

Gameberry Labs

To keep up with the growing number of players, Gameberry decided to host its game solutions on Google Cloud (https://cloud.google.com/) to deliver a consistent gaming experience. Without a dedicated operations team, Gameberry relies heavily on the fully managed infrastructure of Google Cloud to minimize administrative overheads.

“Our ultimate goal in terms of IT infrastructure is to build an online game architecture on Google Cloud that caters to ten million concurrent users, without any developer input,” says Vinesh Battula, Software Developer at Gameberry Labs.

For a brief time from 2017 to 2018, Gameberry hosted its application on a shared hosting platform. This meant that it was sharing resources with other companies. As the number of users grew, this became a challenge because players would encounter frequent game downtime when the server ran out of resources. Gameberry quickly moved to Google Cloud as a cost-effective solution to access dedicated resources for compute, processing, and storage.

Now Gameberry runs multiple instances on Compute Engine (/compute) for its backend and game platform services, including high-performance game servers for Parchisi Star. The company also leverages BigQuery (/bigquery) and Cloud Storage

(/storage) to power up its gaming analytics and Cloud Logging (/logging) to monitor cloud performance.

Gaining operational insights for a smoother gaming experience

Server connection can make or break a player's gaming experience. This is why high-performance dedicated game servers are vital for any game business. If a server is overcrowded, the player may experience a frustrating lag time. Beyond that, the player may lose gameplay and even coins if the server connection drops in the middle of a game.

As player demand spiked for Gameberry in March 2020, Vinesh provided frontline support to keep the game servers running. At the height of COVID-19 lockdowns, the concurrent users for Parchisi Star went up by 50,000 daily. Peak play time for this game tends to be 3 AM in Bangalore, which is 11:30 PM in Madrid, Spain. Vinesh spent many sleepless nights watching app statistics to make sure that none of Gameberry's servers went down.

Gameberry started using Cloud Logging in April 2020 to view logs-based metrics, such as Compute Engine VM and Redis instances. It creates automated alerts for any issues that arise so the support team can resolve problems quickly. For example, if CPU usage exceeds a specific threshold, Cloud Logging will send an email to the support team to add more servers as required.

"Cloud Logging is a significant advantage for us. Instead of waiting around for something to break in the middle of the night, we can analyze log-based metrics and fix problems before they happen," says Vinesh. "By monitoring instance performance, we can keep our games running smoothly for players."

"Every decision we take, from monetization to feature design, is driven by data. In addition to user reviews, BigQuery gives us important insights into player behavior, so we know what they like and what they want to see."

—Vinesh Battula, Software Developer, Gameberry Labs

Speeding up game server deployment with snapshots

Adding game servers used to be a time-consuming process for Gameberry because of interconnected microservices. The company runs front-end game components as individual microservices on the server side. These microservices provide essential game functions such as matching different users before starting a game, tracking the game status of players, and updating player winnings at the end of a game.

Although loosely coupled to accelerate application development, microservices are hard-coded with connection relationships between instances. To add a new server, Vinesh had to carefully change the code and deploy it to every active instance. This manual process can take up to two hours, depending on the traffic.

To automate the release process, Gameberry now uses Compute Engine snapshots to create a new instance from an existing one, complete with installed software, configuration, and application data. This snapshot is saved in Cloud Storage and can be quickly restored when required. Developers use the snapshot to create a new instance in a few simple clicks to add a new game server.

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>
> *—Vinesh Battula, Software Developer, Gameberry Labs*

Uncovering game insights to engage players and drive game

design

As each game grows in popularity, it generates more user data. Once a player launches the game, Gameberry tracks their actions, collecting data that includes everything from their average play time to purchased items. 80 tebibytes of user and game information is now stored on Cloud Storage and streamed to BigQuery for analysis. This data provides valuable insights to understand player behaviors and is used to improve game features.

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Despite the huge workloads, Gameberry can run multiple daily queries in BigQuery, something that would otherwise be impossible to do cost-effectively at that scale if using an in-house data warehouse. This is because the company would need to invest in new servers as the workload escalates. With BigQuery, Gameberry eliminates capital cost by paying only for what it uses.

One of the queries that game developers and product managers look forward to at Gameberry is the daily projection analysis. The company schedules a daily report that shows a complete analysis of the previous day. It uses Looker Studio (https://marketingplatform.google.com/about/data-studio/) to present this data as informative graphs.

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Gameberry's product team also uses BigQuery for deep-dive analysis to plan new releases. BigQuery helps them to understand the impact of a planned feature on app downloads and how players interact with new features and to gain insights on what monetization strategies they can apply.

Leveraging personalization to enhance the real-time gaming experience

Moving forward, Gameberry set its sights on an ambitious goal of 50M daily active users. It also is exploring Google Kubernetes Engine (/kubernetes-engine) to deliver a low latency, real-time seamless experience for its users and further save costs by autoscaling on demand. By testing new features such as exclusive rewards for season pass subscribers and analyzing user behavior with BigQuery, Gameberry will continue to learn more about customer preferences so it can keep refining its game development for greater success.

"With the ability of Google Cloud to scale seamlessly and provide rich insights based on the user data we gather, we will continue to take our games to the next level, building games that excite and entertain players anywhere in the world," says Vinesh.